

Total No. of Questions : 12]

SEAT No. :

P1949

[Total No. of Pages : 4

[5059]-514

B.E. (Civil Engg.) (End Semester)
DAMS AND HYDRAULIC STRUCTURES
(2012 Pattern)

Time : 2 ½ Hours]

[Max. Marks : 70

Instructions to the candidates:-

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8., Q.9 or Q. 10., Q.11 or Q.12.*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right indicate full marks.*
- 4) Use of electronic pocket Calculator is allowed.*
- 5) Assume Suitable data, if necessary.*

- Q1)** a) Discuss the possible social issues associated with dams. [4]
- b) Enlist the types of dams based on hydraulic action and briefly explain any one type with sketch / sketches. [4]

OR

- Q2)** a) State and explain four instruments used for various measurements needed with reference to safety of dams. Draw suitable sketches wherever possible / necessary. [4]
- b) Enlist the types of dams based on the purpose. Explain any one type with sketch. [4]

- Q3)** A gravity dam has base width 60m with upstream face vertical and downstream face with slope 2H:3V.

Knowing that total vertical force above the base is 63MN, overturning moment and resisting moment are 2.2×10^6 kN.m and 3.9×10^6 kNm; determine

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maximum and minimum vertical stresses induced on the foundation. Also find major principal stress at the toe of the dam. Is the dam safe against overturning? [6]

OR

Q4) State the classification of arch dams. Briefly explain the constant angle arch dam with sketch / sketches. When are the arch dams suitable? [6]

Q5) What is meant by 'spillway'? State its purposes. Explain any one spillway with sketch. [6]

OR

Q6) Enlist various spillway gates. Give the classification of hydropower plants based on various criteria (considerations). [6]

Q7) a) Discuss various causes or modes of failure of earthen dams. Draw relevant sketches. [4]

b) Compare Lane's and Bligh's theories of seepage with suitable sketches. [4]

c) An earthen dam of homogeneous material has top width 5m with upstream slope 3H:1V and downstream slope 2H:1V.

Knowing that coefficient of permeability of the dam material is $0.6 \times 10^{-3} \text{ cm/s}$, determine :

i) Phreatic line for the dam, and

ii) Rate of seepage per unit length of dam.

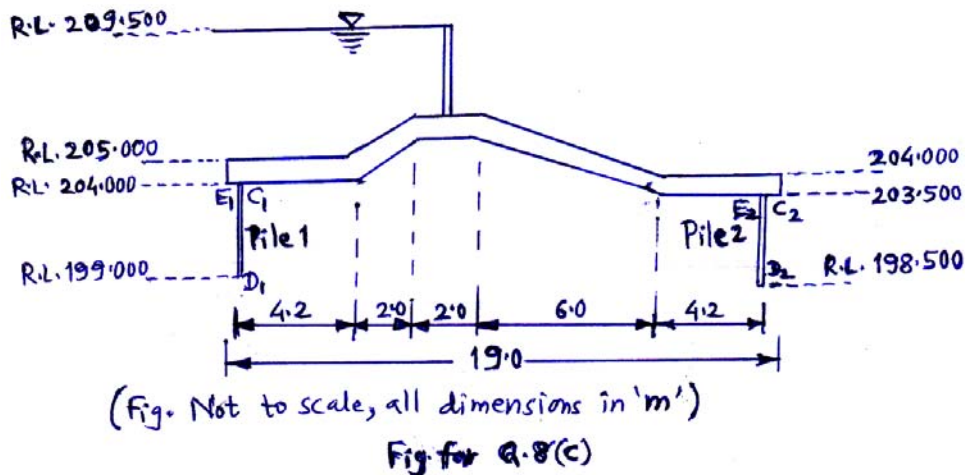
Assume various R.L.s as given below :

- R.L. of deepest river bed = 175 m
- R.L. of top of dam = 201.5 m
- H.F.L. for reservoir = 197 m

For calculating 'y' values, assume interval for 'x' as 15m [10]

OR

- Q8)** a) Discuss the 'Swedish slip circle method' for examining the stability of earth slopes. [4]
- b) Explain one method each for seepage control through [4]
- i) the embankments
- ii) the foundations. (Relevant sketch should be drawn)
- c) Figure shows section of a barrage. R.L.s of key points and horizontal distances are in meters. Calculate % pressure at points D_1 , C_1 , D_2 and E_2 if pressures at E_1 and C_2 are 100% and 0% respectively. Neglect slope corrections. [10]



- Q9)** a) Discuss basic three types of canal alignments with relevant sketch / sketches. [4]
- b) State two requirements of a good canal module (outlet). State four types of canal falls and draw neat sketch of weir type of canal escape. [4]
- c) Design an irrigation canal with side slopes 1H:2V to carry 40 cumecs discharge. Use Kennedy's theory. Assume B:D ratio 2.50, critical velocity ratio (m) = 1 and Kutter's rugosity coefficient = 0.023 Assume $s = \frac{1}{4000}$ for first trial. Take two trials. [8]

OR

Q10) a) State two advantages and two drawbacks / disadvantages of canal lining. [4]

b) What is the necessity of canal falls? Discuss various types of canal outlets (modules) briefly. [4]

c) Design a trapezoidal regime channel to carry a discharge of 50 cumecs using Lacey's theory. Assume side slopes 1H:2V and silt factor $f = 1.10$ [8]

Q11) a) State the types of cross - drainage works. Explain any one C.D. work with suitable sketch. [4]

b) On which factors the choice of C.D. work depends? Briefly discuss. [4]

c) Give classification of rivers based on the topography of river basin. Explain the types briefly. [4]

d) State four objectives of river training. With neat sketch explain briefly the levee (embankment) as a river training work. [4]

OR

Q12) a) With suitable sketches, explain various types of spurs (groynes). [4]

b) Briefly explain three basic types (classes) of river training. [4]

c) Explain Aqueduct and syphon Aqueduct with sketches. [4]

d) What is meant by 'super - passage'? Explain it clearly with sketches. [4]

